

ENGE707 Project Phase 1: Mid-Project Scoping Report

GitHub Repository:

<https://github.com/SrikarKurani/ENGE707-Dr-House>

1. Problem Statement

One of the biggest challenges in healthcare today is predicting outcomes for patients with serious or life-threatening illnesses. Clinicians often need to estimate the likely prognosis of a critically ill patient when deciding on treatment, communicating with patients and their families, and planning for end-of-life care. A wrong prediction could mean the patient is treated unnecessarily, causing suffering, or is under-treated, leading to a loss of life that could have been avoided. Therefore, a data-driven model that could help the clinical judgment providing an estimation of the risk of mortality could be of practical use in the hospital setting, while the final decision would still be up to the healthcare professionals.

The project is to predict the risk of in-hospital mortality for seriously ill patients from clinical, physiological and demographic information collected at or around the time of hospital admission. The primary outcome forecasted is whether a patient dies while in the hospital, denoted by the variable `hospdead`. This outcome was selected in place of the six-month mortality variable (`death`) as in-hospital mortality is more directly relevant to healthcare teams when making treatment and care decisions during a patient's admission.

1.1 Motivation

Uncertainty about prognosis can be stressful for patients and their families, and adds to the pressure on healthcare services, especially in intensive care and emergency settings. A model that could identify patients at higher risk of death earlier could help to enable more informed and timely discussions between clinicians, patients and families about treatment options. This could involve decisions about resuscitation status and whether a move to palliative care may be appropriate.

1.2 Background

SUPPORT (Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments) was a large, multi-centre observational study of seriously ill adults at several major teaching hospitals in the United States (Knaus et al., 1995). It gathered a variety of data including physiologic measures, functional status, demographic information and patient survival. The main goal was to improve the accuracy of outcome predictions and help doctors make better decisions about end-of-life care. The dataset SUPPORT2 we used in this project is a non-identified tabular version of the original study which has been used commonly. The data consists of 9,105 patient records and 47 variables, which is a suitable dataset for developing and evaluating the machine learning model.

1.3 Stakeholders

Clinicians: can use predicted risk estimates to help prioritise patient care and support difficult discussions with patients and their families.

Patients and families: can receive clearer and more consistent information about a patient's likely prognosis, helping them make more informed decisions about treatment and care.

Hospital administrators: can use overall risk information to support planning and allocation of resources, particularly for critical care and palliative care services.

Researchers and policy analysts: can use the dataset to investigate differences in patient outcomes and treatment intensity across different demographic groups, including factors such as income and race.

1.4 Practical Relevance

Machine learning for predicting mortality risk is a well-established use in healthcare and remains of practical importance. Even small improvements in prediction accuracy compared to traditional clinical scoring systems could make a real difference when applied to large numbers of patients. But mortality prediction models present serious ethical and fairness issues, too. For instance, the inclusion of demographic variables like race or income in the model may be problematic as they may be sources of or reinforce existing biases in healthcare.

2. Dataset Description and Justification

The data set is SUPPORT2, a de-identified tabular version of the original SUPPORT study. It is publicly available through Vanderbilt University's Biostatistics teaching materials, and open platforms like UCI and Kaggle. A copy of this project was downloaded from Kaggle on August 1, 2026 (<https://www.kaggle.com/datasets/joebeachcapital/support2>) under a Creative Commons Attribution 4.0 International (CC BY 4.0) license which permits reuse with credit to the original source.

The SUPPORT2 dataset meets all the specified selection criteria for this project:

Tabular, alphanumeric data: The dataset contains 9,105 rows and 47 columns, consisting of numerical and categorical clinical and demographic variables. It does not contain image data.

Feature count: The dataset initially contains 47 features, which is within the required range of 20-300 features.

Sufficient records: With 9,105 patient records, the dataset provides enough data to perform exploratory analysis and to divide the data into training, validation, and testing sets.

Meaningful task: The project uses a binary classification problem, where the model predicts whether a patient dies during their hospital admission or survives. This has clear clinical and practical relevance.

Documentation: The variables are well documented through the original study literature and associated codebook. This provides useful information for understanding and interpreting the variables correctly.

2.1 Variable Groups and Metadata

To support a clear and responsible interpretation of the dataset, the 47 raw variables are grouped below according to their role. Each group includes representative examples and a brief explanation of the type of information captured by the variables.

Group	Example variables	Description
Demographic / socioeconomic	age, sex, race, income, edu	Patient background characteristics recorded at enrolment.
Diagnosis / comorbidity	dzgroup, dzclass, num.co, diabetes, dementia, ca	Primary disease category and count/type of coexisting conditions.
Physiology / labs at admission	meanbp, hrt, resp, temp, crea, sod, bili, alb, wblc, glucose, bun, urine, pafi, ph	Vital signs and laboratory values reflecting the patient's clinical state near admission.
Severity / functional status	scoma, avtisst, sps, aps, adls, adlp, adlsc	Composite illness-severity and functional-status indices.
Outcome / outcome-adjacent	death, hospdead, slos, d.time, surv2m, surv6m, prg2m, prg6m, sfdm2, dnr, dnrday	The modelling target and related fields describing what happened to the patient during or after the admission.

This same grouping was used in Section 4 to identify which variables were suitable to be used as predictors in the model. Demographic, diagnostic, physiological and severity-related variables were considered as possible predictors, while variables directly describing the outcome or events occurring during or after the outcome were classified as outcome-related. Thus, these outcome-related variables were removed from the model inputs to prevent data leakage and to make sure that the model only had access to information that would be available in a real-world setting when making a prediction.

3. Data Acquisition, Inspection and Documentation

We used pandas to load the data set in `python.read_csv`. The output of the notebook shows Table 1, which confirms that the dataset contains 9,105 rows and 47 columns. It has a mix of float64, int64 and object data types which matches with the documented dataset structure. No duplicates were identified.

Missing values were present in 32 of the 47 columns, ranging from under 1% for most physiological variables to much higher levels for `adlp` (61.95%), `urine` (53.40%), `glucose` (49.42%) and `bun` (47.80%). Missingness was higher for laboratory and functional status variables, which may be linked to tests or assessments being conducted only when clinically indicated. Similarly, `income` may be missing as some patients did not report this information. This implies that the missing data are not missing completely at random, which was considered when building the data-cleaning strategy in Section 4.

Eight categorical variables were identified: `sex`, `dzgroup`, `dzclass`, `income`, `race`, `ca`, `dnr`, and `sfdm2`. No obvious invalid or inconsistent categories (e.g. misspelt or duplicate labels) were found. Both `income` and `sfdm2` however had a relatively large proportion of missing values. A decision was therefore made to deal with these directly rather than estimate them.

Also identified were two possible binary outcome variables: `death` (six-month mortality) with a positive rate of 68.1% and `hospdead` (in-hospital mortality) with a positive rate of 25.9%. The target variable for this project was the `hospdead` variable, since in-hospital mortality is more directly related to decisions made during the hospital admission of a patient (see Section 1).

As part of the dataset inspection process, a comprehensive data dictionary mapping all 47 raw columns to plain-text descriptions was developed. This helped to support a clear and responsible interpretation of each variable as described in Section 2.1.

The inspection also identified some physiologically impossible values in the numerical variables, which were not obvious from the summary statistics alone. The variables `meanbp`, `hrt` and `resp` which represent mean arterial blood pressure, heart rate and respiratory rate respectively, all had a few values recorded as exactly zero. `Meanbp` had

51 zero values (0.56% of records), hrt 83 (0.91%) and resp 65 (0.71%). These measurements are more likely to be failed or unrecorded measurements than true zero values, as these measurements would not realistically be zero for a living patient.

Likewise, totmcst and dnrday have negative values that are inconsistent with their definition. There were 2 negative values in totmcst and 195 in dnrday. The minimum values are -\$102.72 and -88 days respectively. These values were therefore identified and recorded as part of the data inspection process. How these values are handled (e.g. whether they are recorded as missing rather than valid zero or negative measurements) is discussed further in Task 3.

4. Data Cleansing and Transformation

4.1 Removal of Leakage and Administrative Columns

Nine columns were dropped before imputation or encoding: death, d.time, slos, surv2m, surv6m, prg2m, prg6m, sfdm2, and hday. This was an important data-quality decision because these variables could leak information that would not be available at prediction time:

death, d.time, slos: These variables have information that is available only after or very close to the patient's outcome. They represent different mortality outcomes, number of days of follow up and total length of hospital stay respectively. Including them would mean the model was using information that would not be available at the moment a real prediction was needed.

surv2m, surv6m, prg2m and prg6m: Probabilities predicted by the original SUPPORT prognostic model or by treating physicians. Correlation analysis showed that they were strongly correlated with hospdead through Pearson correlation values ranging between -0.43 and -0.56. Including these variables could allow the model to mostly reproduce existing clinical predictions rather than learn independently from the raw clinical measurements of the patient. This could make the model appear more accurate than it actually is and would reduce the value of developing an independent prediction model.

sfdm2 and hday: sfdm2 represents the patient's functional status at a follow-up point after admission, so this information would not be available when making the initial prediction. hday is an administrative variable used to identify the study entry day and does not provide useful clinical information for predicting in-hospital mortality.

By removing these nine columns, the dataset was reduced from 47 original columns to 38 potential predictor columns, not including the hospdead target variable. This led to a

feature set that better reflects the information that would realistically be available to clinicians at or near the time of hospital admission.

4.2 Missing Value Handling

Columns with >40% missing values (adlp, urine, glucose, bun) were given a binary `was_missing` indicator before imputation, to maintain the signal that a test may not have been considered necessary. All numerical columns were then imputed with the column median, chosen as it is less sensitive to outliers and more suitable for the heavily right-skewed clinical and cost variables. For categorical variables such as income, race, and DNR, missing values were replaced with an explicit “Missing” category rather than being replaced with the most common category. For example, an unreported income should not automatically be assumed to belong to the most common income bracket. This approach preserves the information that a value was missing, rather than potentially losing that signal during preprocessing.

4.3 Outlier Review

Numeric variables were checked for 1st and 99th percentile. Extreme values (maximum age 101.8 years, hospital charges up to \$1.44 million, and up to nine recorded comorbidities) were considered clinically reasonable for a cohort of seriously ill hospitalised adults rather than data-entry errors. Therefore, they were retained rather than capped or removed. The cost variables (charges, totcst, totmcst) are heavily right skewed and are suggested to be transformed using the `log1p` function before modelling. The exploratory analysis revealed that this transformation has a large normalising effect on the distribution.

4.4 Encoding

The sex variable was coded as a binary variable (male=0, female=1). The income variable is ordinal, so there is a natural order to the categories. So, it was coded with values from 0 to 3, with 0 being an income under \$11k and 3 being an income over \$50k. We created an indicator `income_was_missing` before imputing the missing income values. Missing values were replaced with the median income code. This approach avoids the introduction of an artificial value outside the ordinal range, while preserving information about which values were originally missing.

The rest of the categorical variables, i.e. dzgroup, dzclass, race, ca, dnr etc. do not have a natural ordering. Thus, these variables were one-hot encoded. The first category was omitted for each variable to avoid perfect multicollinearity.

The data cleaning and encoding process is completed and the final transformed data set has 9,105 rows and 58 columns. The cleaned data set was saved at support2_cleaned.csv.

5. Exploratory Data Analysis and Visualisation

This section summarises the main findings from the exploratory data analysis, while the full code and additional outputs are available in the accompanying notebook.

5.1 Target Class Distribution

Figure 3: Target Mortality Distribution Analysis

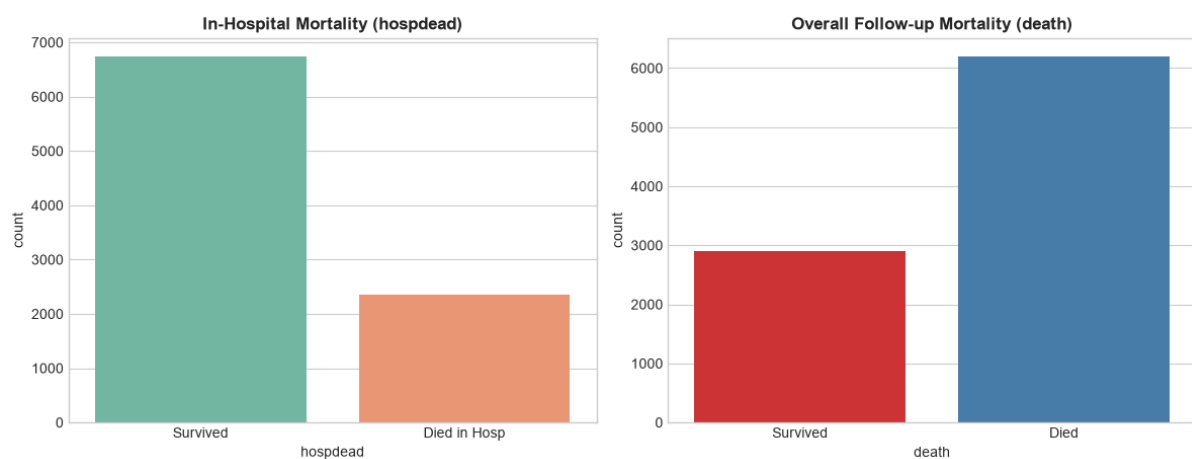


Figure 1: “25.9% of patients died in their hospital admission (hospdead), a moderate class imbalance of around 3:1 in favour of survival. The wider six-month follow-up outcome (death) is far more skewed in the other direction, with 68.1% of patients dead by that point.”

5.2 Missing Value Pattern

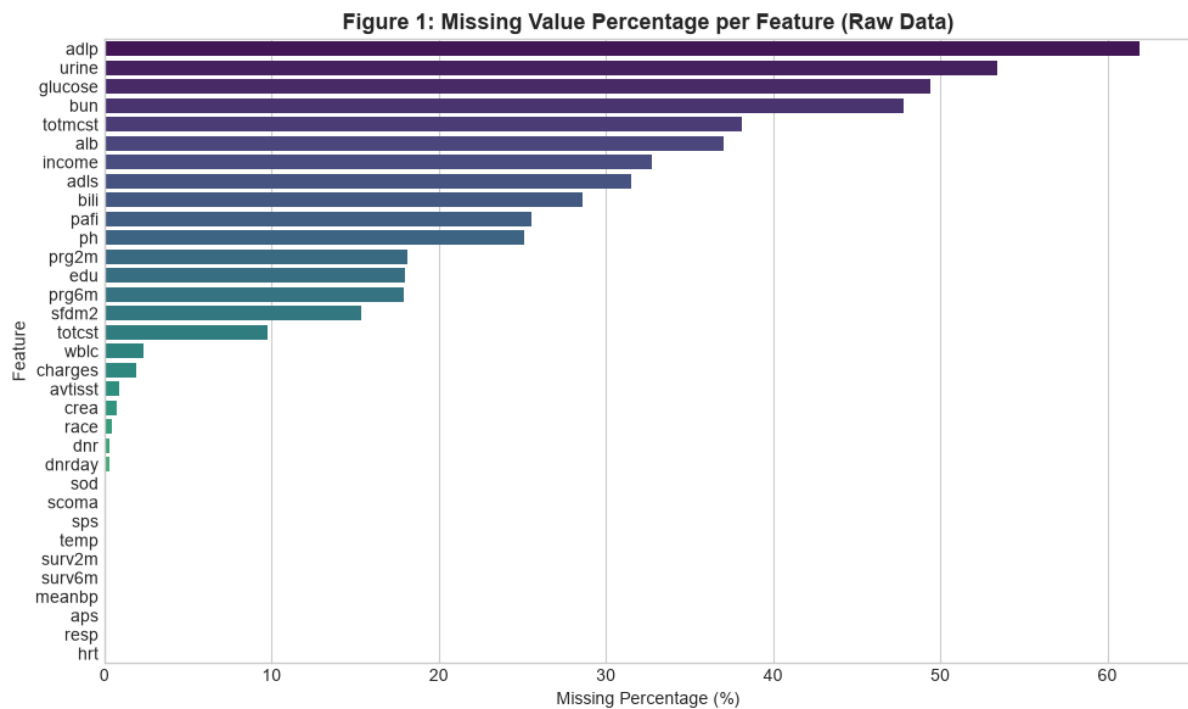


Figure 2: “Missingness is very uneven, ranging from less than 1% for most vital-sign variables to 62% for *adlp*, and is concentrated in optional lab tests and functional status measures.”

5.3 Feature Distributions

Figure 2: Physiological and Demographic Feature Distributions

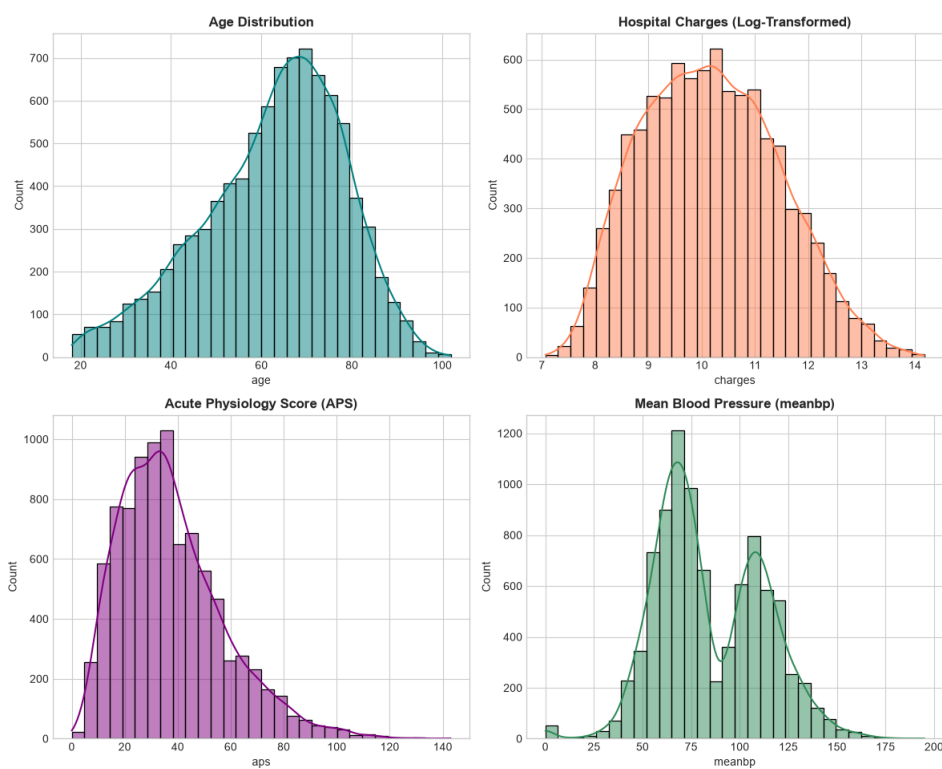


Figure 3: "Age is roughly bell-shaped with a peak in the 60s-70s. Hospital charges $\log_{10}$ transformed are roughly symmetric. Mean blood pressure shows a small spike at zero - the physiologically impossible readings identified in Task 2."

5.4 Mortality by Disease Group

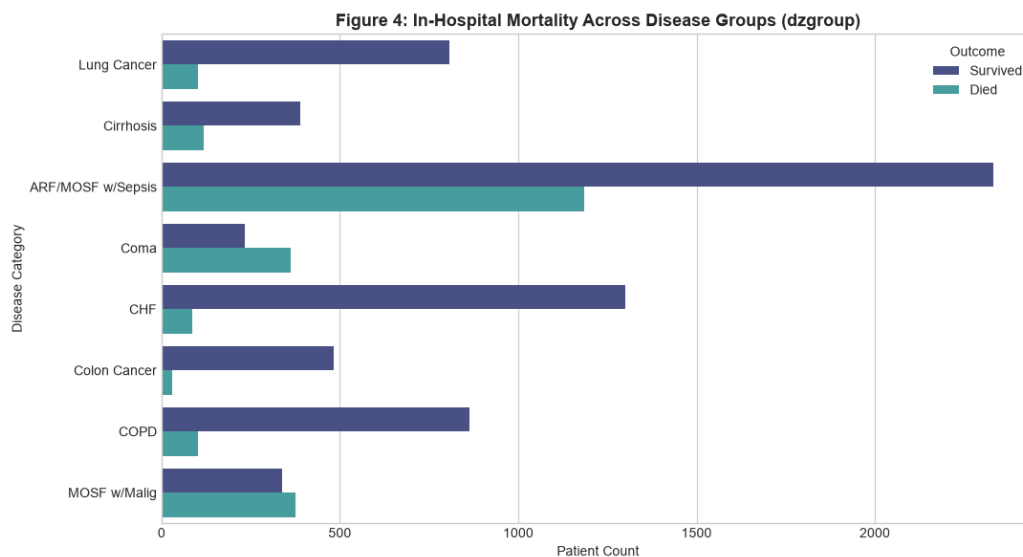


Figure 4: "Death rates vary significantly by diagnosis: Coma (60.7%) and MOSF with Malignancy (52.7%) have the highest in-hospital mortality, while Colon Cancer (5.7%) and CHF (6.3%) have the lowest."

5.5 Correlation Structure

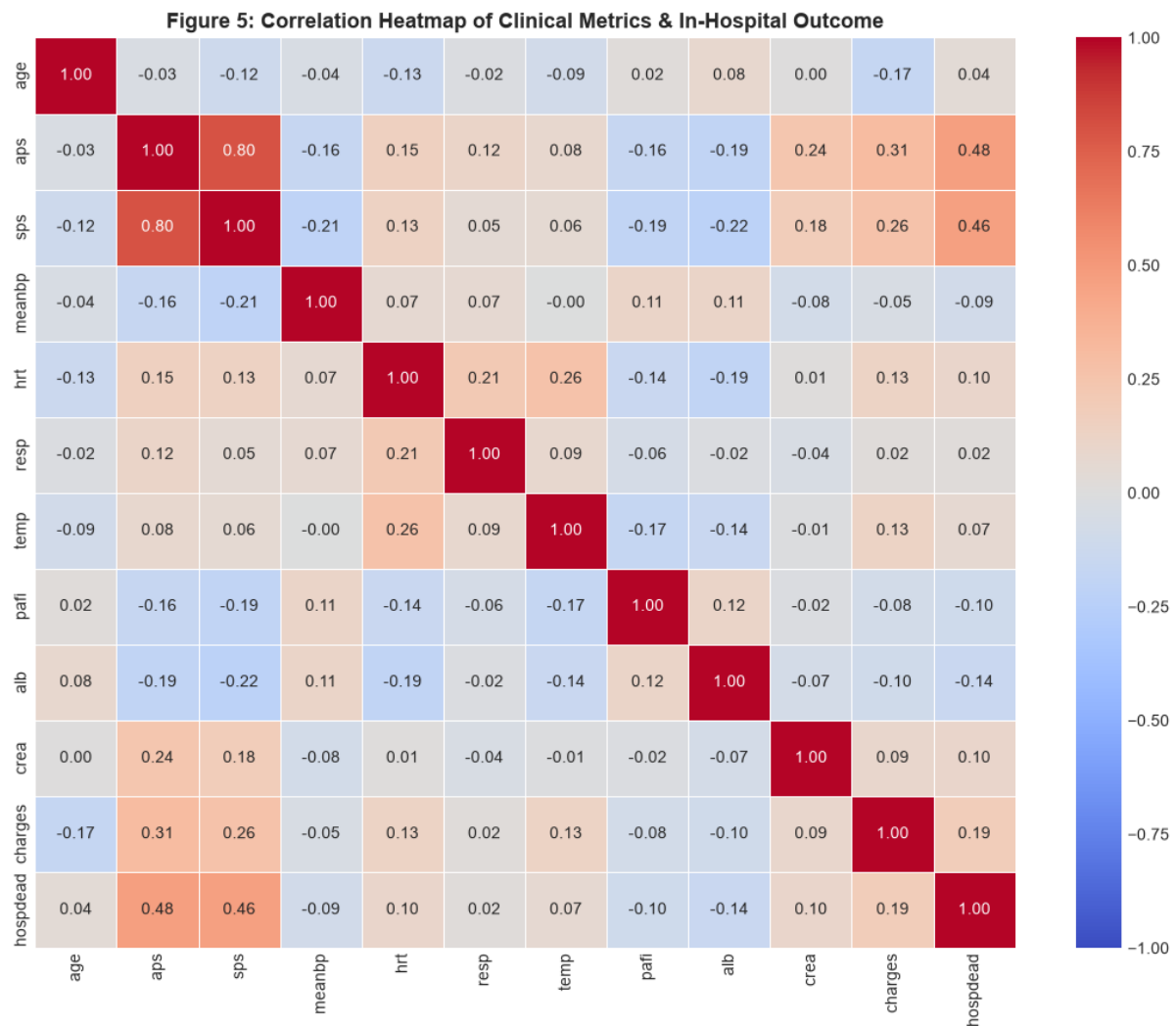


Figure 5: “Composite illness-severity scores were most associated with hospdead: aps ($r=0.48$) and sps ($r=0.46$). Note surv2m and surv6m are intentionally left out here as per Task 3 as they are leakage variables.”

5.6 Physiological Variables by Outcome

Figure 6: Key Physiological Variables by In-Hospital Outcome

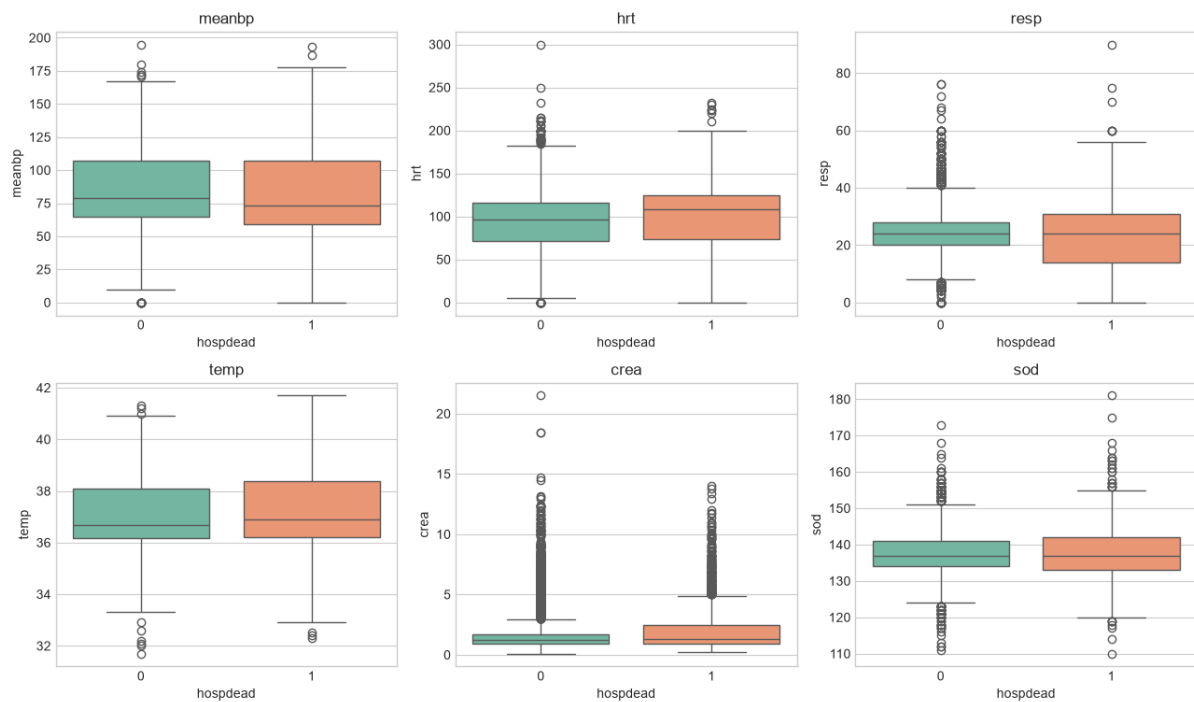


Figure 6: “Patients who died in hospital had a wider spread and slightly lower median blood pressure, along with higher median heart rate and creatinine.”

5.7 Summary of Key Data Insights

To summarise, the exploratory analysis provided several important findings that will directly impact the modelling process in Phase 2:

Class imbalance: About 26% of patients are in the positive class. This can cause accuracy to be a misleading metric for model performance. As such, metrics like precision, recall, F1-score and AUROC should be reported. Approaches such as stratified sampling or class weighting can be used.

Structured missingness: The missing values seem to follow some meaningful pattern and are not completely random. This supports the missing-indicator approach used in Section 4 over simply dropping incomplete records, which would lose a lot of useful data.

Severity scores: Composite severity scores are more strongly associated with the outcome than individual vital sign measurements. Hence, these variables should be considered as important features to use in feature selection and further engineered features could also be investigated.

Physiological variables: There are significant differences between patients who survived and those who died regarding blood pressure, heart rate and respiratory rate. This supports the preservation of these variables as potential predictors in the model.

Disease group variation: Mortality rates change considerably between diagnosis groups, ranging from 5.7% for Colon Cancer to 60.7% for Coma. This indicates that disease group (dzgroup) is likely to be an important predictor of mortality. It also suggests that the model should take into consideration the different risk profiles associated with each diagnosis, rather than treating all patients as having the same level of risk.

Cost-related variables: Many cost-related variables are strongly right-skewed. So a log transformation should be considered before using these with modelling types that are sensitive to skewed distributions.

6. Data Pipeline diagram

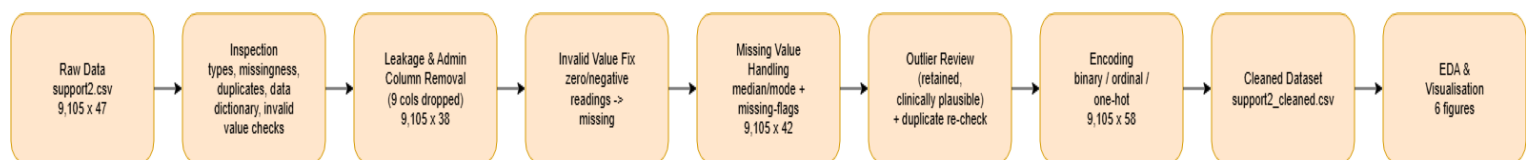


Figure 7: “End-to-end Phase 1 data pipeline, from raw data to a cleaned, EDA-ready dataset. A full-resolution version of this diagram is available in the project repository at [reports/pipeline_diragram.png](#).”

7. Assumptions, Limitations, and Reflection

There are several decisions in this phase worth noting. hospdead was selected as the modeling target rather than death as it better reflects a more immediate clinical decision point but both could be modeled separately in the next phase. One valid strategy is to drop surv2m, surv6m, prg2m and prg6m as leakage, and the other is to employ them as benchmark features to assess the added value of a new model. We have chosen median imputation and an explicit Missing category for their simplicity and reproducibility, but more complex methods (e.g. multiple imputation) could be explored later. The main problem was to remove predictors that were indirectly giving away the answer. Looking at correlations before removing candidate leakage columns was more convincing than just looking at the documentation. A limitation worth noting is that the dataset is mostly white patients from US hospitals in the 1990s, so a model trained on this might not work as well for other populations or in a modern healthcare setting.

8. References

Knaus, W. A., Harrell, F. E., Lynn, J., et al. (1995). The SUPPORT prognostic model: Objective estimates of survival for seriously ill hospitalized adults. *Annals of Internal Medicine*, 122(3), 191-203.

SUPPORT2 dataset. (2023). *Kaggle*.

<https://www.kaggle.com/datasets/joebeachcapital/support2>

Appendix A: Lecturer Approval Evidence

The following approval was received from the ENGE707 Teaching Team confirming the dataset and problem for our Group 'Dr House':

Hi,

Your group's proposed dataset has been approved.

Group name: Dr House

Dataset title: Support2

Dataset link: <https://www.kaggle.com/datasets/joebeachcapital/support2>

You may now proceed with the project.

Regards,

ENGE707 Teaching Team